

SolarPro Global — Technical Guide

Engineering basis, standards, and integration reference

Version: 2026-06-05 (beta) **Audience:** Senior engineers, technical leads, and procurement/IT staff at Ghana installer and supplier companies; internal engineering team. **Companion to:** User Guide (workflow) and Sales Pitch (positioning).

This document explains **how** SolarPro computes its outputs and **which standards** it follows, so an external engineer can defend any value the platform produces to a client or to a financing bank.

1. Engineering principles in one page

SolarPro is built on three rules:

1. **Every output is derivable from the inputs.** No "magic" multipliers — every coefficient is listed in this guide or in **assumptions.md**.
2. **Every output is overridable.** Defaults are conservative; the engineer can replace any value with their own.
3. **Every output is traceable.** The proposal PDF shows the calculation path so a reviewing engineer can validate without re-running the tool.

2. Standards applied

Standard	Where it's applied	Notes
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BS 7671 (18th Edition, +A2)	AC cable sizing, derating, protective device coordination	UK origin; default standard for Ghana commercial/industrial wiring
IEC 62548	DC string design, MPPT voltage windows, isolation	International PV-array standard
IEC 60364-7-712	PV-specific installation requirements	Used for protective bonding, isolation
IEC 61730	Module safety classification	Determines Class II isolation requirements
NEC Article 690	Fallback for US-style installations	Triggered only when site country = USA
Ghana Energy Commission — Solar PV Code of Practice (2019)	Permitting checklist on the installation report	Drives the EC-certification section
IEC 61724	Performance ratio & yield calculation	Drives the economic analysis

Standards are cited inline in every report so the bank's engineer can verify.

3. Site assessment

3.1 Irradiance

Source: `config/global_solar_data.py` — region-level peak sun hours (PSH) lookup.

For Ghana:

Region	PSH (kWh/m ² /day)	Ambient (°C)
Greater Accra	5.2	28
Ashanti	5.0	26

Northern	5.6	30
Volta	5.1	27
Western	4.8	26

Source data: NASA POWER + Ghana Meteorological Agency monthly means, averaged 2018–2023.

3.2 ECG tariff

Region-tagged residential and commercial tariffs sourced from Public Utilities Regulatory Commission (PURC) quarterly publications. Escalation default = 6%/year (BoG inflation forecast).

The engineer can override tariff and escalation in the economic step.

4. Load analysis

$$\text{Daily energy demand (Wh/day)} = \sum (P_i \times n_i \times h_i \times d_i / 7)$$

Where:

- P_i = nameplate power of load i (W)
- n_i = quantity
- h_i = hours per day
- d_i = days per week (default 7)

A diversity factor is applied to the **peak** load (not the daily energy):

$$\text{Peak (kW)} = \sum (P_i \times n_i) \times DF$$

Where DF = 0.65 for residential, 0.85 for commercial, 1.0 for industrial. The engineer can change DF per project.

Surge load = peak $\times$ 3 (motor start; configurable 2–5 $\times$).

5. PV array sizing

5.1 Required peak power

$$P_{pv} \text{ (kWp)} = E_{\text{daily}} / (PSH \times \eta_{\text{system}} \times \delta_{\text{temp}})$$

Where:

- E_{daily} = daily energy demand (kWh)
- PSH = peak sun hours from §3.1
- η_{system} = 0.80 (default) — combined inverter + wiring + soiling
- $\delta_{\text{temp}} = 1 - \gamma \times (T_{\text{cell}} - 25)$ — temperature derate

Cell temperature: $T_{\text{cell}} = T_{\text{amb}} + 25 \text{ }^{\circ}\text{C}$ (NOCT approximation; configurable). γ (power temperature coefficient) = -0.0034 K^{-1} (Tier-1 mono panel default).

5.2 Module count

$$N_{\text{modules}} = \text{ceil}(P_{pv} \times 1000 / W_{\text{module}})$$

Where $W_{\text{module}} \in \{110, 250, 330, 400, 450, 500, 550\} \text{ Wp}$ (commit [27b2492](#)).

5.3 String design

For each candidate inverter MPPT range $[V_{\text{min}}, V_{\text{max}}]$:

$$\begin{aligned} N_{\text{series_min}} &= \text{ceil}(V_{\text{min}} / (V_{\text{mp}} \times \delta_{T_{\text{hot}}})) \\ N_{\text{series_max}} &= \text{floor}(V_{\text{max}} / (V_{\text{oc}} \times \delta_{T_{\text{cold}}})) \end{aligned}$$

Temperature corrections:

- $\delta_{T_{\text{hot}}} = 1 + \beta_{\text{voc}} \times (T_{\text{cell_hot}} - 25) \approx 0.88$ (V_{mp} shrinks when panel is hot)
- $\delta_{T_{\text{cold}}} = 1 + \beta_{\text{voc}} \times (T_{\text{cold}} - 25) \approx 1.10$ (V_{oc} grows when panel is cold)

$\beta_{\text{voc}} = -0.0029 \text{ K}^{-1}$ default.

If `N_series_min > N_series_max`, the MPPT panel is incompatible → engine flags WARNING and suggests a smaller-Vmp module or a different inverter.

5.4 Mounting derate

Mounting type	Soiling factor	Temperature uplift (°C)
<code>rooftop_pitched</code>	0.95	+5
<code>rooftop_flat</code>	0.90	+10
<code>rooftop_metal</code>	0.95	+12
<code>rooftop_membrane</code>	0.95	+8
<code>ground_fixed</code>	0.92	+3
<code>ground_tracking</code>	0.95	+3

These factors fold into `η_system` and `T_cell` automatically.

6. Battery sizing

$$C_{\text{battery}} \text{ (Wh)} = E_{\text{daily}} \times N_{\text{autonomy}} / (\text{DoD} \times \eta_{\text{inv}} \times \eta_{\text{batt}})$$

Defaults:

- `N_autonomy` = 1 day (residential), 0.5 day (grid-tied), 2 days (off-grid critical)
- `DoD` = 0.80 (lithium), 0.50 (gel/AGM)
- `η_inv` = 0.95
- `η_batt` = 0.95 (lithium), 0.85 (lead acid)

The battery type (lithium / gel / AGM) drives DoD and `η_batt`. The engineer chooses; the engine reflects the math.

Output is rounded up to the nearest stocked battery module size (5 kWh, 10 kWh, 15 kWh, 20

kWh by default — configurable in Settings → ROQ)

7. Inverter sizing

$$P_{inv} \text{ (kW)} = \max(\text{Peak_kW} \times 1.25, P_{pv} \times 0.85)$$

Two-sided constraint:

- Must handle peak load with 25% margin
- Must absorb at least 85% of PV array output (Ghana standard avoids over-clipping under low PSH)

A hybrid inverter is selected by default for off-grid + battery; grid-tied selects a string inverter.

8. Cable sizing

8.1 DC cable (panel → combiner → inverter)

Voltage drop method (IEC 62548):

$$A \text{ (mm}^2\text{)} = (2 \times L \times I \times \rho) / \Delta V_{\max}$$

Where:

- L = one-way length (m)
- I = string current at STC
- ρ = 0.0175 $\Omega \cdot \text{mm}^2/\text{m}$ (Cu @ 20 °C)

- $\Delta V_{\max} = 1.5\% \times V_{\text{array}}$

Round up to the next standard size (4, 6, 10, 16, 25 mm²). Verify the result against ampacity at 70 °C ambient — bump up one size if borderline.

8.2 AC cable (inverter → DB → load)

Per BS 7671 §523, Table 4D5 (PVC SWA):

Cable	Current carrying capacity (A) @ 30 °C ambient
2.5 mm ²	24 A
4 mm ²	32 A
6 mm ²	41 A
10 mm ²	57 A
16 mm ²	76 A
25 mm ²	101 A

Derating applied:

- Ambient temperature factor (BS 7671 Table 4B1) — Ghana ambient 32–35 °C → 0.94
- Grouping factor (Table 4C1) — if more than one cable shares conduit
- Installation method factor (clipped direct vs. conduit)

Final: smallest size whose derated capacity $\geq$ design current $\times$ 1.25.

Voltage drop check: $\leq$ 3% for final circuits, $\leq$ 5% combined (per BS 7671 Appendix 12).

8.3 Earthing

Earth conductor sized per BS 7671 §544 — half the phase conductor's cross-section, minimum 4 mm² Cu.

Earth rod: 1.2 m $\times$ 16 mm Cu — measured resistance $\leq$ 10 Ω per IEC 60364-7-712.

9. Bill of Quantities — how it's built

`calc_boq()` traverses the sized system and emits one line per material category:

Category	Quantity logic
Modules	N_{modules} from §5.2
Mounting rails	$\text{ceil}(N_{\text{modules}} \times 1.05 \text{ m} / 4.2 \text{ m})$ — 4.2 m standard rail
End clamps	$4 \times N_{\text{strings}}$
Mid clamps	$2 \times (N_{\text{modules}} - N_{\text{strings}})$
MC4 connectors	$2 \times N_{\text{strings}}$
DC cable	length from §8.1 with 10% wastage
AC cable	length from §8.2 with 10% wastage
Inverter	N_{inv}
Battery modules	$\text{ceil}(C_{\text{battery}} / W_{\text{battery_unit}})$
Earth rod	1 per system
Earth conductor	length matches AC cable
Surge arrestor (DC)	1 per MPPT input
Surge arrestor (AC)	1 per phase
Isolators (DC)	1 per string
Combiner box	1 per inverter
Ground mount only	steel post (1 per 4 modules), concrete footing (1 per post), purlin beam (4.2 m per row), earth rod (1 per 20 modules)

Pricing:

- Default: blank — engineer types unit cost.
- With supplier integration enabled: pulls live pricing via the supplier's API. Currently

10 Economic analysis

10.1 Energy yield (per IEC 61724)

$$E_{\text{year}} \text{ (kWh)} = P_{\text{pv}} \times \text{PSH} \times 365 \times \text{PR}$$

PR (performance ratio) defaults:

- Rooftop residential: 0.75
- Rooftop commercial: 0.80
- Ground mount fixed: 0.82
- Ground mount tracking: 0.86

10.2 Cash flow

For year $y \in [1, \text{lifetime}]$:

$$\begin{aligned} \text{Savings}_y &= E_{\text{year}} \times (1 - \delta_y) \times \text{Tariff} \times (1 + \text{escalation})^{(y - 1)} \\ \text{OPEX}_y &= \text{OPEX}_0 \times (1 + 0.03)^{(y - 1)} \\ \text{Net}_y &= \text{Savings}_y - \text{OPEX}_y \\ \text{NPV}_y &= \text{Net}_y / (1 + \text{discount})^y \end{aligned}$$

Where:

- δ_y = degradation per year (default 0.5%, lithium battery system; 0.7% pure PV)
- OPEX_0 = annual maintenance estimate (default 1% of CAPEX)

10.3 Headline metrics

- **Payback (years)** — interpolated to the year where cumulative cashflow ≥ 0
- **IRR (%)** — Newton-Raphson on the cashflow vector, tolerance 1e-6
- **NPV (GHS)** — sum of NPV_y over lifetime
- **LCOE (GHS/kWh)** — $\sum \text{Costs} / \sum \text{Energy}$

11. Reports

Every project produces seven PDF reports, all rendered from the same JSON state:

1. **PV Sizing Report** — array, strings, MPPT match, temperature checks
2. **BOQ** — full materials list with prices
3. **Cable Schedule** — every run with gauge, length, derating, voltage drop
4. **Economic Analysis** — charts + IRR/NPV table
5. **Installation Method Statement** — step-by-step, with safety
6. **Energy Yield Report** — month-by-month projection
7. **Bankable Proposal** — combined cover-page + executive summary + sized excerpts of 1–6

Reports embed the assumptions used (PSH, system efficiency, derating factors) so a third-party engineer can validate without re-running the tool.

12. Data model (for IT staff)

12.1 Storage

Each project's full engineering state lives in a single JSON blob: `projects.data_json`. Schema (abbreviated):

```
{
  "site": { "country": "Ghana", "region": "Greater Accra", "psh": 5.2, "ambient": 30.0 },
  "loads": [ { "name": "Fridge", "watts": 250, "qty": 1, "hours": 12, "days_per_year": 365 },
  "totals": { "daily_kwh": 18.3, "peak_kw": 4.1, "surge_kw": 12.3 },
  "results": {
    "pv": { "p_kwp": 6.0, "n_modules": 12, "w_module": 500, "strings": [ ... ] },
    "battery": { "kwh": 10, "type": "lithium" },
    "inverter": { "kw": 5.0, "type": "hybrid" },
    "cable": { "dc_mm2": 6, "ac_mm2": 4 },
    "boq": [ ... ],
    "economics": { "payback_y": 4.2, "irr": 0.27, "npv_25y": 86_200 }
  }
}
```

12.2 Export

- **PDF** — **Generate Proposal** button on the project page.
- **JSON** — **GET /api/project/<id>/export.json** (authenticated; returns the schema above).
- **CSV** — **GET /api/project/<id>/boq.csv** (BOQ only).

12.3 Re-import

POST a previously exported JSON to **/api/project/import** — recreates the project. Useful for restoring from backup or moving a project between accounts.

13. Integration points

For supplier and installer companies, three integration surfaces are available during beta:

13.1 Pricing API (suppliers)

If you sell panels / inverters / batteries / cables, we'll wire your stock into the BOQ procurement screen. Required from you:

- A REST or GraphQL endpoint returning JSON with { **sku**, **name**, **category**, **unit_price_ghs**, **in_stock**, **lead_time_days** }
- Optional webhook for stock updates

We pull on a 6-hour cadence (configurable). Live unit costs replace blank cells in the BOQ.

13.2 Project handoff (installers)

After a designer signs off a project, a **/installer/handoff** webhook fires to your endpoint with the full JSON state. Your project management system picks it up and creates a work order.

13.3 Single sign-on (SSO)

If your company uses Google Workspace, Microsoft 365, or any OIDC provider — we support SSO on Business+ plans. Required: client ID + client secret + issuer URL.

14. Security & compliance

Control	Implementation
Authentication	Username + password; 5-failed-attempt logout 15 min
Session management	HttpOnly Lax cookie; CSRF (<code>_csrf</code>) on every POST
Database isolation	Per-tenant row-level filters on all user data tables
Encryption at rest	Hosting platform-managed (currently AES-256)
Encryption in transit	TLS 1.2+ end-to-end
Audit log	<code>audit.log</code> (JSON-line) — login, project edits, exports
Security headers	CSP, X-Content-Type-Options, Referrer-Policy, Frame-Ancestors
Data export	Self-serve from Settings → Export (per GDPR Art. 20)
Data deletion	Self-serve from Settings → Delete Account
Backups	Daily snapshot, 30-day retention

Reference: `SECURITY.md` in the repo for the full Zero-Trust architecture + RBAC matrix.

15. Performance & SLA targets (beta)

Metric	Target	Notes
Sizing calculation latency	< 30 s	p95
Proposal PDF generation	< 60 s	p95

Optimic	99.9%	Free Railway tier, will rise to 99.9% on RGS
Support response (in-app chat)	< 60 s	Rule-based; AI fallback to Claude/Llama
Support response (email)	24 h	Business hours

16. Where to look in the code (internal staff)

Concern	File
Loads → sizing pipeline	<code>web_app.py</code> → <code>calc_loads</code> / <code>calc_pv</code> / <code>calc_battery</code>
AC cable sizing	<code>calculation/ac_cable_sizing.py</code>
DC cable sizing	<code>web_app.py</code> → <code>size_dc_cable</code>
BOQ generation	<code>web_app.py</code> → <code>calc_boq</code>
Economic engine	<code>web_app.py</code> → <code>calc_economics</code>
Global irradiance data	<code>config/global_solar_data.py</code>
AI helpline	<code>web_app.py</code> → <code>/api/assistant/chat</code> + <code>api_manager.py</code>
Email send chain	<code>api_manager.py</code> → <code>_send_email</code>
Structured logging	<code>logging_config/structured_logger.py</code>
Health endpoints	<code>web_app.py</code> → <code>/api/health*</code>

17. Roadmap (next 90 days)

Month	Theme	Highlights
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June 2026	Data hardening	Custom domain SSL, multi-tenant edge isolation, screenshots in user guide
July 2026	Procurement	Nocheski + Ozo pricing live in BOQ, supplier directory page
2026	Energy reporting	Month-by-month yield, Excel export of BOQ, mobile-optimised dashboard

Feature requests during beta: send to support@aiappinvent.com or via in-app chat. Top-3 by demand each fortnight ships on the next sprint.

18. Engineering changelog (June 17, 2026)

Four mechanical changes shipped to master today. The first three concern the shading dashboard's mathematical contract; the fourth introduces a mount-aware electrical drawing.

18.1 Sun-position Bézier — single source of truth

The shading dashboard's central viewport draws a dashed yellow sun-path arc as the SVG path

```
M 60 460 Q 500 -340 940 460
```

a quadratic Bézier with horizon endpoints (60, 460) and (940, 460) and control point (500, -340) so the noon peak sits at (500, 60). The same Bézier coefficients now drive the JS animation:

```
function sunXY(hour) {
  var t = (hour - 6) / 12;
  var x = (1 - t) * (1 - t) * 60 + 2 * (1 - t) * t * 500 + t * t * 940;
  var y = (1 - t) * (1 - t) * 460 + 2 * (1 - t) * t * (-340) + t * t * 460;
  return { x: x, y: y };
}
```

with `BASE_X = 700`, `BASE_Y = 70` matching the server-rendered `_sun_x` / `_sun_y` defaults, so the disk's transform `translate(p.x - BASE_X, p.y - BASE_Y)` puts it at `(p.x, p.y)` for every hour. Five visible marker circles sit on the curve at `t = 1/12, 3.5/12, 6/12, 8.5/12, 11/12` (07:00, 09:30, 12:00, 14:30, 17:00) — pre-computed Jinja constants so they never drift.

Net: the visible curve and the moving disk are guaranteed to share a single mathematical formula.

Any client asking "is your simulation accurate?" can be shown both the SVG path string and the JS coefficients side by side.

18.2 Shading-factor precedence chain

The dashboard now reads `_factor` / `_label` / `_loss` from a single ordered precedence:

1. `shading.factor` (the saved value written by `_apply_shading_factor`)
2. `_eng.bucket_factor` (cached engine block, used only when no save exists)
3. `1.00 / ' ' / 0` (no-data defaults)

`_apply_shading_factor` (in `web_app.py`) runs the deterministic geometry engine first, falls back to the heuristic agent `_compute_shading_factor` when project context is incomplete, and yields to a manual operator override when the gold-pill Save handler fires. All three write the same `data["shading"]["factor"]` field, so the dashboard now displays the actually-applied value regardless of which source produced it.

When `factor_source == "manual"`, a `· MANUAL` suffix is appended to the displayed label, and the AGENT PICK row highlight in the SHADING_FACTORS table tracks the saved factor rather than the cached engine bucket. The engine block remains in `data["shading"]["engine"]` so the 3D scene still has geometry to render against.

18.3 Engine ↔ heuristic snapping parity

Both `engine.shading_engine.SHADING_BUCKETS` and `web_app.SHADING_FACTORS` are the same eight rows (No shading → Very severe shading, 0% → 40% loss, 1.00 → 0.60 factor) with the same conservative snap rule: pick the highest row whose loss% is $\leq$ the computed loss%. When either path runs alone its output is one of those eight rows, so the visible factor on the dashboard is one of eight discrete values — never a continuous figure. Downstream code (loads / sizing / BOQ) reads only the snapped row.

18.4 Directed sun-rays — wide-beam cone-of-light rendering

The shading viewport's sun-rays used to live inside `#svgSunGroup` and translate with the disk, which made them look frozen relative to the array. They have been moved into a sibling `<g id="svgDirectedRays">` group OUTSIDE the sun group, and a new `paintDirectedRays(hour)` is wired into the `paintSun` wrapper chain. Per slider tick the group's `innerHTML` is replaced with a fresh build.

Per illuminated target the renderer stacks three layers in z-order:

1. An OUTER cone polygon `<polygon fill="#fde68a" opacity="0.18">` — apex at the current sun emit point `(p.x, p.y + 22)`, base on the target's plane with half-width 90 px for the PV array and a per-obstruction clamped `Math.max(22, Math.min(bw * 0.55, 60))` for each obstruction.
2. An INNER hot core polygon `<polygon fill="#fef3c7" opacity="0.22">` at 45 % of the outer half-width.
3. A fan of ray accents `<line stroke="#fbbf24">` on top — 9 lines across the array at $\pm 90 / \pm 67 / \pm 45 / \pm 22 / 0$ px (centre 1.8 px @ 0.65 opacity, outer 1.0 px @ 0.38 opacity) and 5 lines per obstruction at relative offsets $-1.0 / -0.5 / 0 / +0.5 / +1.0 \times$ that target's beam half-width.

Obstruction geometry is read live from each `[id^="shadow-"]` polygon's `data-base-x / data-base-w / data-h` attributes, so beams stay in sync with whatever obstruction set the project carries. The group is cleared (`innerHTML = ''`) when `hour < 5 || hour > 19` to mirror the existing `paintShadows` night cutoff. Implementation lives in `templates/shading.html`, function `paintDirectedRays`, immediately above the `paintSun` wrapper.

18.5 Collateral build pipeline — PDFs, TTS audio, paired-image video

`build_collateral_pdfs.py` is now a three-stage pipeline rather than just a PDF build. Stage 1 renders each `docs/src/*.md` source to a branded PDF via `markdown-pdf` (the same `MarkdownPdf` + `Section` flow we've used since the beta). Stage 2 calls `edge-tts` with anonymous Microsoft Edge endpoints (`en-US-AriaNeural` for the user walkthrough, `en-US-GuyNeural` for the technical walkthrough) to produce zero-cost neural-voice MP3s — see [\[\[feedback-zero-cost-apis\]\]](#). Stage 3 composes each MP3 with a paired screenshot via the `ffmpeg` binary bundled by `imageio-ffmpeg` (no system install required) into a 1280×720 MP4 with letterbox padding and `-tune stillimage` for tiny file sizes. The `/support/asset/<slug>` route exposes the MP4s under the `video/mp4` MIME, replacing the earlier MP3 entries in the `_SUPPORT_ASSETS` whitelist.

18.6 Installation drawings — mount-aware string routing (Drawing 1B)

The Installation Drawings report (`/project/<pid>/report/installation/drawings`) carries a new Drawing 1B between the existing PV panel internal wiring diagram and the battery bank diagram. One SVG with three Jinja branches keyed on the project's stored `mounting_type`:

Mount class	Topology rendered	Standards cited

rooftop_sloped (default, pitched, hip, gable, metal, rooftop_pitched, rooftop_metal)	Cables clipped to under-tray, EPDM-flashed pitched penetration, IP65 rigid conduit to indoor inverter wall	BS 7671 Method B-1, 0.85 thermal-insulation factor; §712 PV bonding
rooftop_flat (flat, membrane, concrete, rooftop_flat, rooftop_membrane)	Galvanised cable tray on ballast stands, parapet transition to UV-rated rigid conduit, drop to plant room	IEC 60364-5-52 tray fill ≤ 40 %, 70 °C PVC; FM 4474 ballast wind-load
ground_* (ground_fixed, ground_tracking)	IP67 armoured conduit buried ≥ 600 mm, cable draw-pits every 90 m, combiner on equipment shelter	BS 7671 Tab 4D4A buried-cable derating; BS 7499 array frame bonding

The diagram is sized from the project's actual **pps** (panels per string) × **num_strings**, colours each string from an 8-step palette (red → orange → yellow → green → blue → violet → magenta → cyan), and draws DC+ red / DC– blue cable paths from the array to the combiner box (fuse + DC isolator per string, IP65 housing, "{N}-in / 1-out") and out to the inverter (MPPT input ceiling annotated). The notes panel adds string-balance, derating, and combiner-termination guidance with BS 7671 §537 lockout requirement.

Implementation: see **templates/report_installation_drawings.html** block "DRAWING 1B" (≈ 150 LOC SVG + 3-column notes panel). All **mt** branches are derived from the lower-cased **mounting_type** string, so any new raw mount value the form might add ("ground_pole", "rooftop_carport", etc.) defaults to the closest-class topology rather than 500'ing.

19. Screenshots index

The technical guide references the same screens as the user guide:

- **[SCREEN: location-globe]** — for §3 site assessment
- **[SCREEN: loads-input]** — for §4 load analysis
- **[SCREEN: results-pv]** — for §5 PV sizing
- **[SCREEN: boq]** — for §9 BOQ
- **[SCREEN: economics]** — for §10 economics
- **[SCREEN: installation-drawings]** — for §6.4 mounting derate consequences

